ARCHIT KUMAR

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PROFESSIONAL SUMMARY

Computer Science graduate (VIT, May 2026) with hands-on experience building production-grade AI systems across GenAI, machine learning, and computer vision. Developed RAG pipelines, agentic workflows, and ML-backed APIs using LangChain, FastAPI, PyTorch, and vector databases. Promoted to Team Lead of a 10-member benchmark-evaluation team while maintaining an 87% task approval rate; focused on reliability through fallback handling, confidence-based escalation, and structured output validation.

EDUCATION

Vellore Institute of Technology Sep 2022 - May 2026

Bachelor of Technology in Computer Science, GPA: 8.45 *Vellore, Tamil Nadu*

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, C++
LLM & Agentic AI: LLM Evaluation, Benchmark Engineering, Agentic Workflows, Agent Trajectory Analysis, Rubric Design, Prompt Engineering, LangChain
Machine Learning: PyTorch, scikit-learn, XGBoost, Random Forest, SHAP
NLP & Computer Vision: Hugging Face Transformers, spaCy, NLTK, Vision Transformer, ResNet, Tesseract OCR
Databases & Vector Search: PostgreSQL, MySQL, Redis, FAISS
Backend & Tools: FastAPI, REST APIs, SQLAlchemy, Docker, Linux, Bash, Git/GitHub, Cursor, Postman, React

PROFESSIONAL EXPERIENCE

SDE Intern — Agent Evaluation & Benchmark Engineering May 2026 - Oct 2026

Xelron.ai *Remote*

- Authored and validated hard, verifiable benchmark tasks for frontier coding agents (GPT-5.5/Codex, Claude Opus) across ML, security, systems administration, and build-tooling domains — task specifications, deterministic behavioral tests, Dockerized environments, oracle solutions, and anti-cheat safeguards — maintaining an 87% approval rate across 4 project workstreams.
- Promoted to Team Lead within two months, leading a 10-member task-authoring team through technical reviews, agent-failure trajectory analysis, and difficulty calibration to keep submissions within the target solve-rate band.
- Designed rubric-scored, multi-step evaluation tasks spanning coding-agent benchmarks and domain-specific occupational scenarios, building realistic input datasets, gold-standard reference solutions, and weighted scoring rubrics validated against model outputs.
- Used Cursor, Claude, and GPT-based tools daily for repository analysis, debugging, and implementation planning, independently verifying submissions via CI checks, containerized test harnesses, and repeated agent evaluation runs.

GenAI Intern (WatsonX Training Program) May 2025- July 2025

IBM *Remote*

- Implemented Retrieval-Augmented Generation (RAG) pipelines using WatsonX foundation models for document-based Q&A and summarization.
- Built an AI chatbot on IBM Watson Assistant, applying prompt engineering to design conversational flows and structured prompts for text classification, summarization, and code generation.

PROJECTS

GenAI-Powered Multi-Modal Assistant | *Python, Streamlit, LangChain, OpenAI, FAISS* | **GitHub** | **Demo**

- Built a production-ready AI assistant with 5 autonomous workflows (research agent, document Q&A, code generation, content creation, support triage) handling multi-format inputs via LangChain (RAG + FAISS).
- Developed LangChain and FAISS RAG pipelines for PDF, DOCX, image, and URL inputs, combining retrieval, OCR, entity extraction, and grounded question answering.
- Improved reliability with citation checks and critique-revision for research outputs, syntax verification with one-pass repair for generated code, and confidence-based escalation.

Climate Risk Prediction System | *Python, PyTorch, ViT, ResNet, XGBoost, FastAPI, React* | **GitHub** | **Demo**

- Built an end-to-end climate risk platform analyzing satellite imagery using a EuroSAT-trained Vision Transformer, ResNet, and XGBoost ensemble, achieving ~87% classification accuracy.
- Developed a FastAPI backend with Celery and Redis for asynchronous image analysis and used Socket.io events for real-time dashboard updates after processing.
- Added PostgreSQL/pgvector similarity search for historical analyses and LangChain/Gemini mitigation reports with fallback handling; delivered results through a React geospatial dashboard.

AI Doctor Assistant (Disease Prediction System) | *Python, FastAPI, Random Forest, SHAP, Gemini, FAISS, Docker* | **GitHub**

- Built a Dockerized FastAPI application for preliminary symptom-based disease prediction using a Random Forest model (~85% accuracy), with SHAP explanations highlighting important symptoms.
- Created a Gemini 2.0-assisted triage flow combining follow-up questioning with a local Hugging Face/FAISS retrieval layer for precaution guidance.
- Implemented a lab-report pipeline using Tesseract OCR and pdfplumber for text extraction, Gemini Vision for structured analysis, and regex fallback when API rate limits occur.

LEADERSHIP & ACTIVITIES

Core Member July 2023 - Apr 2024

AI/ML Club

- Led a team of 15 members organizing 20+ technical workshops and hackathons reaching 500+ students.